

Customer Segmentation Dashboard Using RFM Analysis

Project Overview

This project analyzes customer purchasing behavior using the **RFM** (Recency, Frequency, Monetary) model and presents the insights through an interactive Power BI dashboard. The dashboard helps businesses identify valuable customers, understand customer segments, and make data-driven marketing decisions.

Objectives

- Analyze customer purchasing patterns.
 - Segment customers using RFM Analysis.
 - Identify high-value and low-value customers.
 - Build an interactive dashboard for business insights.
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Dataset

- **Dataset:** Online Retail Dataset
 - **Records:** ~540,000 transactions
 - **Features Used:**
 - CustomerID
 - InvoiceDate
 - Quantity
 - UnitPrice
 - Country
 - InvoiceNo
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Tools & Technologies

- Power BI Desktop
 - Power Query
 - DAX
 - Microsoft Excel
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Data Cleaning

- Removed missing Customer IDs.
 - Removed duplicate records.
 - Removed cancelled/invalid transactions.
 - Filtered negative or zero quantities.
 - Created **Total Price** column.
 - Corrected data types.
 - Extracted date from InvoiceDate.
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RFM Analysis

Recency (R)

Days since the customer's last purchase.

Frequency (F)

Total number of purchases.

Monetary (M)

Total amount spent.

Customers were scored from **1--5** for each metric and grouped into business-friendly customer segments.

Customer Segments

-  Champions
-  Loyal Customers
-  Potential Loyalists
-   At Risk
-  Lost Customers

Dashboard Features

- Total Customers KPI
- Total Revenue KPI
- Average Frequency KPI
- Average Customer Value KPI
- Customer Segment Distribution
- Revenue by Customer Segment
- Top Customers by Revenue
- R Score Distribution
- F Score Distribution
- M Score Distribution
- Customer Segment Filter
- Revenue Range Slicer

Key Insights

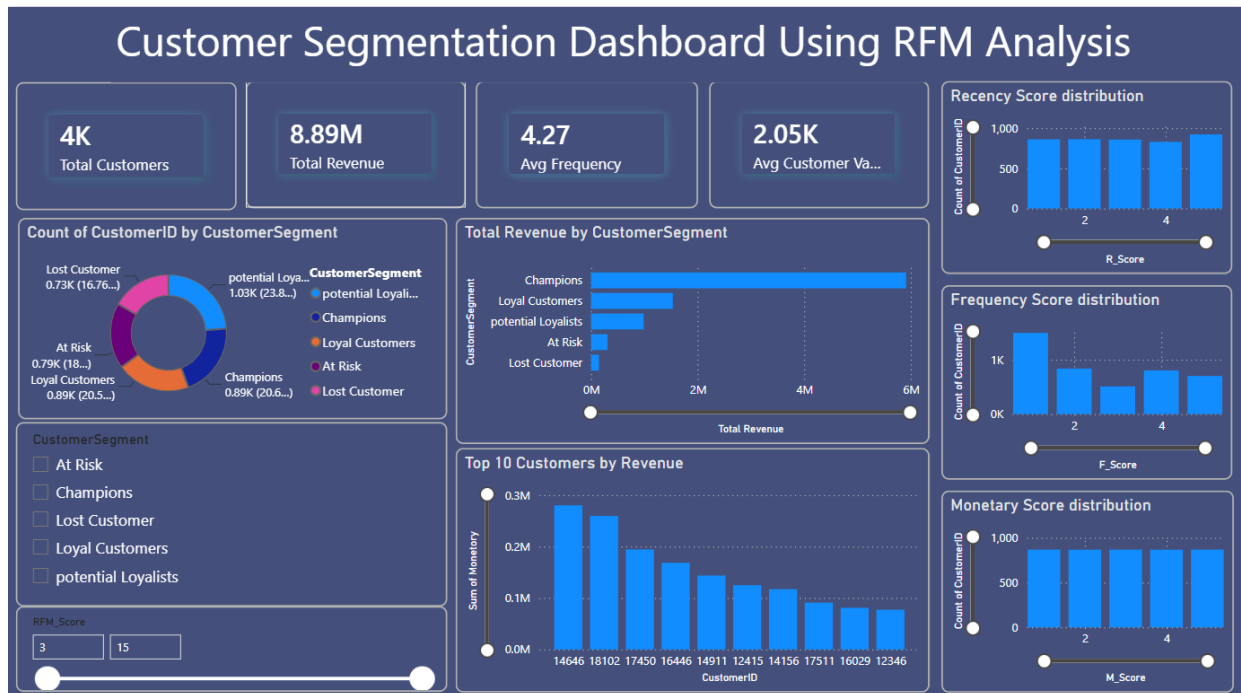
- Champions generated the highest revenue.
- Loyal customers contributed significantly to sales.
- At Risk and Lost customers can be targeted with retention campaigns.
- RFM segmentation enables personalized marketing strategies.

Skills Demonstrated

- Data Cleaning
- Power Query

- DAX
- RFM Analysis
- Data Visualization
- Dashboard Design
- Business Intelligence

📊 Dashboard Preview



👤 Author

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